

Knot Theory

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MA498 Project I

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to the

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CERTIFICATE

This is to certify that the work contained in this project report entitled “Title of the project report” submitted by Srinjoy Som (Roll No.: 220123074) and Chaitanya Chhabra (Roll No.: 220123012) to the Department of Mathematics, Indian Institute of Technology Guwahati towards partial requirement of Bachelor of Technology in Mathematics and Computing has been carried out by them under my supervision.

It is also certified that this report is a survey work based on the references in the bibliography. It is also certified that, along with literature survey, a few computational implementations have been carried out and simulation studies have been done by the students under the project.

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ABSTRACT

Knot theory examines the embedding of closed curves in three-dimensional space and their equivalence under continuous deformation. This report introduces fundamental analytical tools including knot diagrams and Reidemeister moves, which provide a systematic framework for distinguishing distinct knot types. We explore computational variants such as polygonal and lattice knots, which discretize the embedding space to enable algorithmic analysis. Particular attention is given to lattice knots in $\mathbb{Z}^3$, where recent research has focused on determining minimal edge requirements for knot formation and developing computational methods for knot equivalence testing. The discrete structure of lattice knots facilitates both theoretical bounds and exhaustive computational enumeration, making them valuable for both mathematical analysis and applications in statistical mechanics and polymer science.

Our algorithmic and computational work on lattice knots is available at <https://github.com/srinjoysom11/Latticeknots>.

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Chapter 1

Introduction

1.1 What is a Knot?

Informally, a knot may be thought of as a closed piece of string embedded in three-dimensional space such that the string does not intersect itself. The ends of the string are joined together to form a closed loop. The central problem of knot theory is to determine when two such loops are essentially the same, that is, when one can be smoothly deformed into the other without cutting or passing the string through itself.

Formally, a **knot** is defined as a homeomorphic image of a circle in three-dimensional Euclidean space. That is, a knot is a subset of $\mathbb{R}^3$ that can be represented as the image of an embedding

$$K : S^1 \hookrightarrow \mathbb{R}^3,$$

where S^1 denotes the unit circle.

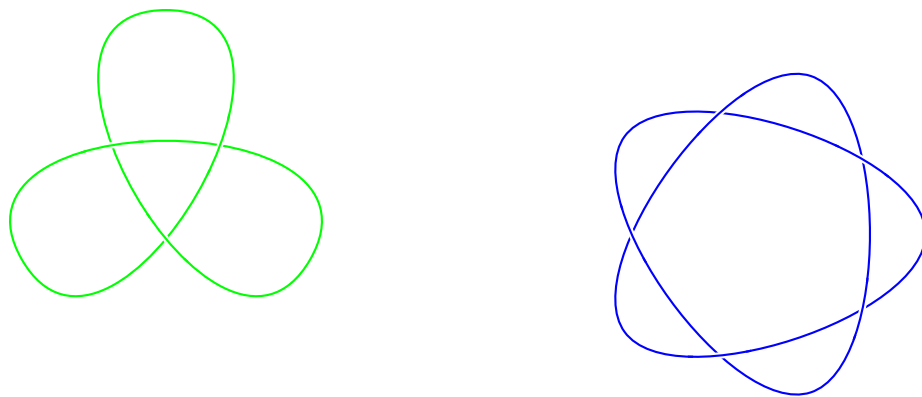


Figure 1.1: Trefoil(green) and Cinquefoil(blue) knot

1.2 Knot Equivalence

Two knots K_1 and K_2 in $\mathbb{R}^3$ are said to be *equivalent* if there exists a homeomorphism $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $h(K_1) = K_2$. This clearly defines an equivalence relation on the set of knots.

A knot that is equivalent to the standard unit circle

$$\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = 1, z = 0\}$$

is called a *trivial knot* (or *unknot*). The equivalence classes of knots in this relation are called *knot types*. There is also a closely related concept known as *ambient isotopy*, which has stronger criteria than equivalence. It is not always true that we can move a knot in space to convert into another knot that is equivalent to it. We shall discuss ambient isotopy in detail later and explain its relation to the above definition of knot equivalence.

1.3 Tame and Polygonal Knots

A polygonal knot is the union of a finite number of closed straight-line segments called edges, whose endpoints are the vertices of the knot. A knot is tame if it is equivalent to a polygonal knot; otherwise, it is wild. In other words, a tame knot can be represented as a closed polygonal curve in $\mathbb{R}^3$ without self-intersections. The behaviour of wild knots might be very counterintuitive and confusing, causing knot theorists to mostly focus on tame knots.

Polygonal knots are convenient both conceptually and computationally. Furthermore, every smooth knot (that is, one described by a continuously differentiable embedding) is equivalent to a polygonal knot.

1.4 Knot Ambient Isotopy

In our earlier discussion, we defined two knots K_1 and K_2 in $\mathbb{R}^3$ to be *equivalent* if there exists a homeomorphism $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $h(K_1) = K_2$.

An *ambient isotopy* of $\mathbb{R}^3$ is a continuous family of homeomorphisms

$$\{h_t : \mathbb{R}^3 \rightarrow \mathbb{R}^3 \mid 0 \leq t \leq 1\}$$

such that:

1. h_0 is the identity map on $\mathbb{R}^3$, i.e., $h_0(p) = p$ for all $p \in \mathbb{R}^3$;
2. the function $H : [0, 1] \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $H(t, p) = h_t(p)$ is continuous .

The parameter t can be interpreted as *time*, so that for each fixed point $p \in \mathbb{R}^3$, the path $h_t(p)$ traces the motion of that point as the entire space (and hence the knot) is deformed over time.

Two knots K_1 and K_2 are said to be *ambient isotopic* if there exists an ambient isotopy $\{h_t\}$ of $\mathbb{R}^3$ such that

$$h_1(K_1) = K_2.$$

In this case, we say that K_1 and K_2 belong to the same *isotopy type*. Intuitively, one may imagine the space $\mathbb{R}^3$ as an elastic medium; an ambient isotopy corresponds to continuously deforming this medium so that the knot embedded within it changes shape, but not its essential topological character. It is important to note that isotopy is an equivalence relation: it is reflexive (any knot is isotopic to itself), symmetric (if K_1 is isotopic to K_2 , then K_2 is isotopic to K_1), and transitive (if K_1 is isotopic to K_2 and K_2 to K_3 , then K_1 is isotopic to K_3). The set of all knots in $\mathbb{R}^3$ is therefore divided into equivalence classes, each representing a unique ambient isotopy type. Knot equivalence is a weaker relation than isotopy. If two knots are isotopic, they have to be equivalent, but the converse is not true.

Orientation and Ambient Isotopy

Every homeomorphism of $\mathbb{R}^3$ is either *orientation-preserving* or *orientation-reversing*. Intuitively, a homeomorphism preserves orientation if it maps every right-handed screw to a right-handed screw, and reverses orientation if it transforms a right-handed screw into a left-handed one. The identity map is orientation-preserving, while the reflection

$$(x, y, z) \mapsto (x, y, -z)$$

is orientation-reversing. Since an ambient isotopy begins with the identity map and depends continuously on t , it follows that every h_t in the isotopy must be orientation-preserving. Consequently, a necessary condition for two knots to be ambient isotopic is

the existence of an *orientation-preserving* homeomorphism of $\mathbb{R}^3$ mapping one onto the other.

Amphicheiral Knots

A knot K is said to be *amphicheiral* (or *achiral*) if there exists an orientation-reversing homeomorphism $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $h(K) = K$. Equivalently, a knot is amphicheiral if it can be mapped onto its mirror image K^* under some orientation-preserving homeomorphism of $\mathbb{R}^3$.

For example, the figure-eight knot is amphicheiral — it can be deformed continuously into its mirror image — while the trefoil knot (or cloverleaf knot) is not.

1.5 Knot Projection

A *knot projection* is a way to represent a knot in $\mathbb{R}^3$ by projecting it onto a plane. Formally, it is the image of the knot under a regular projection $p : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, such that at most two distinct points of the knot project to the same point in the plane, and the set of such points (called *crossings*) is finite.

At each crossing, one strand of the projection passes *over* the other. By indicating which strand passes over and which passes under, we obtain a *knot diagram*. Thus, a knot diagram is a planar representation of a knot that retains all the necessary topological information to recover the original three-dimensional knot.

Knot projections and diagrams play a fundamental role in knot theory, as they allow us to study and classify knots through two-dimensional drawings. Later, we will see that through a sequence of local moves called *Reidemeister moves*, one can transform one diagram of a knot into another equivalent one, without changing the knot type.

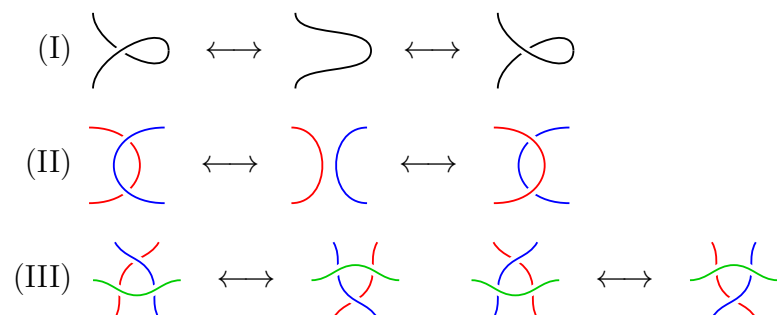
1.6 Reidemeister Moves

When a knot in $\mathbb{R}^3$ is projected onto the plane, the resulting knot diagram provides a combinatorial representation of the knot. However, different projections of the same knot may yield diagrams that appear quite different. To determine when two diagrams represent the same knot type, we rely on a fundamental set of diagrammatic operations

known as the **Reidemeister moves**.

A *Reidemeister move* is a local modification of a knot diagram that alters only a small neighbourhood of the diagram, leaving the rest unchanged. There are exactly three such moves:

1. **Type I:** This move introduces or removes a single loop in the diagram. Geometrically, it corresponds to twisting the knot in three-dimensional space without changing its overall topology. It affects the crossing number by ± 1 .
2. **Type II:** This move introduces or removes a pair of crossings. One strand is pushed over another, creating or eliminating two crossings simultaneously. This move preserves the parity of the crossing number.
3. **Type III:** This move slides one strand over a crossing between two other strands. No crossings are created or removed; instead, the relative ordering of crossings is changed. This move is essential for preserving the structure of multi-crossing regions in diagrams.



Each move is confined to a small disc in the plane, and outside this disc the diagram remains unchanged. These moves provide the basic operations by which one can transform one diagram into another without altering the underlying knot type.

Reidemeister moves play a fundamental role in knot theory for several reasons:

- they allow knots to be studied using planar diagrams rather than full 3D embeddings;
- they form the basis for defining many knot invariants (e.g., Jones polynomial, writhe, linking number);
- they give a complete combinatorial description of isotopy for tame knots.

There are numerous examples where two diagrams that look very different can be shown to represent the same knot through a sequence of such moves.

Theorem 1.6.1 (Reidemeister Theorem). *Two diagrams of the same polygonal knot represent isotopic knots in $\mathbb{R}^3$ if and only if one diagram can be transformed into the other by a finite sequence of Reidemeister moves.*

Summary

A knot is an embedding of the circle in three-dimensional space, studied up to isotopy. We have clarified that only tame knots—those equivalent to polygonal or smooth embeddings—are considered, as they correspond to the physical and computationally tractable cases. The concept of isotopy formalises the intuitive idea of deforming a knot without cutting or self-intersection and establishes the framework for defining and studying knot equivalence. These definitions provide the foundation for the further development of knot theory in both algebraic and geometric directions.

Chapter 2

The Fundamental Group

The purpose of this chapter is to introduce the notion of the fundamental group, an algebraic invariant that encodes the way in which loops behave under continuous deformation within a topological space. We begin with basic notions from topology, building the machinery required to define paths, their concatenation, and the equivalence of paths under homotopy.

2.1 Topological Preliminaries

Definition 2.1.1. A *topological space* $(X, \mathcal{T})$ is a pair consisting of a set X and a collection $\mathcal{T}$ of subsets of X , called *open sets*, satisfying the following axioms:

1. $\emptyset, X \in \mathcal{T}$,
2. the union of any collection of members of $\mathcal{T}$ is again in $\mathcal{T}$,
3. the intersection of any finite collection of members of $\mathcal{T}$ is again in $\mathcal{T}$.

The elements of $\mathcal{T}$ are the open sets of X , and the pair $(X, \mathcal{T})$ is said to be endowed with the topology $\mathcal{T}$.

Remark 2.1.2. The topology $\mathcal{T}$ determines the notion of *closeness* in X without reference to a metric. The complements of open sets are called *closed sets*, and the collection of closed sets also satisfies axioms dual to those of open sets: arbitrary intersections and finite unions of closed sets are closed.

Definition 2.1.3. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A map $f : X \rightarrow Y$ is said to be *continuous* if for every open set U in Y , the preimage $f^{-1}(U)$ is open in X . Equivalently, f is continuous if and only if for every point $x \in X$ and every neighborhood V of $f(x)$, there exists a neighborhood U of x such that $f(U) \subseteq V$.

Remark 2.1.4. This definition generalizes the familiar notion of continuity from metric spaces to arbitrary topological spaces. It ensures that the image of points under f varies without abrupt jumps with respect to the topology.

Definition 2.1.5. A bijective map $f : X \rightarrow Y$ between topological spaces is a *homeomorphism* if both f and its inverse f^{-1} are continuous. In this case, we write $X \cong Y$ and say that X and Y are *homeomorphic*.

Theorem 2.1.6. *The relation $\cong$ of homeomorphism between topological spaces is an equivalence relation on the class of all topological spaces.*

Proof. We verify the three properties:

1. *Reflexivity:* The identity map $\text{id}_X : X \rightarrow X$ is continuous and bijective, and its inverse is itself, hence a homeomorphism.
2. *Symmetry:* If $f : X \rightarrow Y$ is a homeomorphism, then $f^{-1} : Y \rightarrow X$ is continuous and bijective, hence a homeomorphism.
3. *Transitivity:* If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are homeomorphisms, then their composition $g \circ f : X \rightarrow Z$ is continuous and bijective, with inverse $f^{-1} \circ g^{-1}$ continuous. Hence $g \circ f$ is a homeomorphism.

Thus, homeomorphism defines an equivalence relation on topological spaces. □

Remark 2.1.7. A homeomorphism preserves all topological properties of a space, including connectedness, compactness, and path-connectedness. Hence, spaces that are homeomorphic are topologically indistinguishable—they differ only by a continuous deformation.

Definition 2.1.8. Let $(X, \mathcal{T})$ be a topological space. A *path* in X is a continuous function

$$a : [0, \|a\|] \rightarrow X,$$

where $\|a\| \geq 0$ is the *length of the parameter interval* (the *stopping time* of a in Crowell–Fox). The point $a(0)$ is called the *initial point* and $a(\|a\|)$ the *terminal point*. If $a(0) = a(\|a\|)$, the path is called a *loop*.

Remark 2.1.9. When convenient, the domain of a path may be rescaled linearly to $[0, 1]$ without changing the path up to reparametrization. Paths and loops are the basic objects in algebraic topology through which one studies the structure of spaces via continuous deformation (homotopy). The subsequent development of the fundamental group and groupoid will formalize this idea.

2.2 Paths and Loops

The concept of a path formalizes the idea of moving continuously within a topological space. Paths are the basic objects used to study connectedness, homotopy, and, ultimately, the algebraic structure encoded by the fundamental group.

In this section, we follow the convention established in Section 2.1.8: a path has a domain $[0, \|a\|]$ where $\|a\| \geq 0$ is the stopping time. This allows identity paths to be represented naturally as having domain $[0, 0]$, and enables a coherent treatment of path concatenation with general stopping times.

Definition 2.2.1. Let X be a topological space. A *path* in X is a continuous function

$$a : [0, \|a\|] \rightarrow X$$

where $\|a\| \geq 0$ is the *stopping time* of the path. The point $a(0)$ is called the *initial point* of the path, and $a(\|a\|)$ is called its *terminal point*. For a path from p to q , we write $a : p \rightarrow q$ or $a \in \Omega(X; p, q)$.

Intuitively, a path traces a continuous motion from $a(0)$ to $a(\|a\|)$ within the space X . For example, in $\mathbb{R}^2$, the map $a(t) = (1 - t)x_0 + tx_1$ on $[0, 1]$ gives a straight-line path from x_0 to x_1 with stopping time $\|a\| = 1$. An identity path at a point p has domain $[0, 0]$ and maps the single point 0 to p .

Definition 2.2.2. A *loop* in X based at a point $p \in X$ is a path

$$a : [0, \|a\|] \rightarrow X$$

such that $a(0) = a(\|a\|) = p$. The point p is called the *basepoint* of the loop. A loop has stopping time $\|a\| > 0$ unless it is the identity loop (the degenerate case $\|a\| = 0$).

Loops can be thought of as closed journeys that start and end at the same location. The collection of all loops based at p plays a central role in algebraic topology, as it forms the foundation of the fundamental group $\pi_1(X, p)$.

Definition 2.2.3. If $a : [0, \|a\|] \rightarrow X$ and $b : [0, \|b\|] \rightarrow X$ are paths such that the terminal point of a equals the initial point of b , i.e. $a(\|a\|) = b(0)$, their *product* or *concatenation* is the path $a \cdot b : [0, \|a\| + \|b\|] \rightarrow X$ defined by

$$(a \cdot b)(t) = \begin{cases} a(t), & 0 \leq t \leq \|a\|, \\ b(t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|. \end{cases}$$

The stopping time of $a \cdot b$ is thus $\|a \cdot b\| = \|a\| + \|b\|$.

This operation corresponds to traversing path a first and then immediately following it by path b . The direct concatenation (without reparametrization) ensures that the domain of the product is the union of the two domains, with their lengths additive. This naturally gives identity paths domain $[0, 0]$, which act as genuine identity elements.

Remark 2.2.4. Path multiplication is strictly associative as defined: if a, b, c are composable paths, then

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

as functions. This is a key advantage of the general stopping time convention over the reparametrized $[0, 1]$ convention.

The identity path $e_p : [0, 0] \rightarrow X$ at a point p , defined by $e_p(0) = p$ (with domain the singleton $\{0\}$), acts as a genuine identity element under concatenation:

$$a \cdot e_{a(\|a\|)} = a = e_{a(0)} \cdot a.$$

This makes the algebraic structure much cleaner than when using reparametrization.

Definition 2.2.5. Given a path $a : [0, \|a\|] \rightarrow X$, its *inverse path* $a^{-1} : [0, \|a\|] \rightarrow X$ is the path defined by

$$a^{-1}(t) = a(\|a\| - t).$$

The inverse path has the same stopping time as the original: $\|a^{-1}\| = \|a\|$.

The path a^{-1} traverses the same trajectory as a , but in the opposite direction—from $a(1)$ back to $a(0)$. When considering loops, a^{-1} represents retracing the same route backwards to the basepoint.

Remark 2.2.6. These notions together—paths, loops, concatenation, and inverses—form the algebraic backbone of the study of *path homotopy classes*. Under suitable equivalence, the set of loops based at a point p becomes a group, called the *fundamental group* of X at p , denoted $\pi_1(X, p)$.

2.3 Homotopy of Paths

While paths capture the idea of continuous motion within a space, the notion of *homotopy* allows us to formally express when two such motions can be continuously deformed into one another. This concept lies at the heart of algebraic topology—it allows us to classify paths not by their specific trajectories, but by their essential “shape” or behavior within the space.

Definition 2.3.1. Let $a, b : [0, \|a\|] \rightarrow X$ be paths in a topological space X having the same initial and terminal points and the same stopping time $\|a\| = \|b\|$. A *homotopy between a and b relative to endpoints* is a continuous map

$$H : [0, 1] \times [0, \|a\|] \rightarrow X$$

such that

$$\begin{aligned} H(0, t) &= a(t), & H(1, t) &= b(t), \quad \forall t \in [0, \|a\|], \\ H(s, 0) &= a(0) = b(0), & H(s, \|a\|) &= a(\|a\|) = b(\|a\|), \quad \forall s \in [0, 1]. \end{aligned}$$

We then say that a and b are *path-homotopic relative to endpoints*, and write $a \simeq b$.

Intuitively, H provides a continuous family of paths $\{H(s, \cdot)\}_{s \in [0, 1]}$ deforming a into b . For each fixed s , the map $t \mapsto H(s, t)$ represents a path in X with stopping time $\|a\|$, while the parameter s indicates the “time” during which the deformation occurs. The

conditions on the endpoints ensure that throughout the deformation, the start and end points remain fixed—only the intermediate points of the path may move.

Example 2.3.2. In $\mathbb{R}^2$, consider two paths a and b from $(0, 0)$ to $(1, 0)$, both with stopping time $\|a\| = \|b\| = 1$:

$$a(t) = (t, 0), \quad b(t) = (t, \tfrac{1}{2} \sin(\pi t)).$$

Then

$$H(s, t) = (t, s \cdot \tfrac{1}{2} \sin(\pi t))$$

defines a homotopy from a to b on $[0, 1] \times [0, 1]$. Here s continuously “flattens” the sine curve into the straight line segment. This illustrates the intuitive idea that in a convex space like $\mathbb{R}^2$, all paths with the same endpoints and stopping time are homotopic relative to endpoints.

Theorem 2.3.3. *The relation $\simeq$ is an equivalence relation on the set of all paths in X having the same endpoints.*

Proof. **Reflexivity:** For any path $a : [0, \|a\|] \rightarrow X$, define $H(s, t) = a(t)$ for all $(s, t) \in [0, 1] \times [0, \|a\|]$. This is a constant homotopy from a to itself.

Symmetry: Suppose $a \simeq b$ via $H : [0, 1] \times [0, \|a\|] \rightarrow X$. Then define

$$\tilde{H}(s, t) = H(1 - s, t).$$

This reverses the deformation parameter s , providing a homotopy from b back to a .

Transitivity: Suppose $a \simeq b$ via H_1 and $b \simeq c$ via H_2 , where all three paths have the same stopping time $\|a\| = \|b\| = \|c\|$. Define a new map

$$H(s, t) = \begin{cases} H_1(2s, t), & 0 \leq s \leq \frac{1}{2}, \\ H_2(2s - 1, t), & \frac{1}{2} \leq s \leq 1. \end{cases}$$

Since $H_1(1, t) = H_2(0, t) = b(t)$, this defines a continuous concatenation of the two homotopies, producing a homotopy from a to c . $\square$

Remark 2.3.4. The equivalence classes of paths under $\simeq$ (relative to endpoints) are called *path-homotopy classes*. In many spaces, there are infinitely many distinct paths between

two points, but when considered up to homotopy, these often collapse into a smaller and more meaningful set.

For instance, in a convex subset of $\mathbb{R}^n$, any two paths with the same endpoints are homotopic, so there is only one path-homotopy class between any two points. In contrast, in spaces with “holes,” like the punctured plane $\mathbb{R}^2 \setminus \{0\}$, different ways of winding around the hole correspond to distinct homotopy classes. This phenomenon is what gives rise to nontrivial fundamental groups.

Remark 2.3.5. The path-homotopy relation preserves the structure of path concatenation. That is, if $a_1 \simeq a_2$ and $b_1 \simeq b_2$ with $a_i(1) = b_i(0)$ for $i = 1, 2$, then

$$a_1 \cdot b_1 \simeq a_2 \cdot b_2.$$

This compatibility is essential in defining a well-behaved group operation on equivalence classes of loops, leading naturally to the construction of the *fundamental group* $\pi_1(X, p)$.

2.4 The Fundamental Groupoid $\Gamma(X)$

The fundamental groupoid generalizes the notion of the fundamental group by keeping track of all possible basepoints rather than fixing one. Its elements correspond to homotopy classes of paths with arbitrary stopping times, with composition defined when the terminal point of one path coincides with the initial point of another.

Definition 2.4.1. Let X be a topological space. The *fundamental groupoid* $\Gamma(X)$ is defined as follows:

- The *objects* of $\Gamma(X)$ are the points of X .
- For any two points $p, q \in X$, the *morphisms* from p to q are the path-homotopy classes of paths $a : [0, \|a\|] \rightarrow X$ with $a(0) = p$ and $a(\|a\|) = q$. We denote this set by $\Gamma(X)(p, q)$.
- If $[a] \in \Gamma(X)(p, q)$ and $[b] \in \Gamma(X)(q, r)$, then the composition is defined by

$$[a] \cdot [b] := [a \cdot b],$$

where the concatenated path $(a \cdot b) : [0, \|a\| + \|b\|] \rightarrow X$ is given by

$$(a \cdot b)(t) = \begin{cases} a(t), & 0 \leq t \leq \|a\|, \\ b(t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|. \end{cases}$$

Theorem 2.4.2. $\Gamma(X)$ is a groupoid; that is, for every $p, q, r \in X$,

1. composition is well-defined on homotopy classes,
2. composition is strictly associative,
3. for each $p \in X$, the constant path $e_p : [0, 0] \rightarrow X$ (with $e_p(0) = p$) acts as an identity,
4. every morphism $[a] \in \Gamma(X)(p, q)$ has an inverse $[a^{-1}] \in \Gamma(X)(q, p)$.

Proof. We verify each property in turn.

(1) Well-definedness. Suppose $a_1 \simeq a_2$ and $b_1 \simeq b_2$ (relative to endpoints), with $a_i(\|a_i\|) = b_i(0)$ for $i = 1, 2$, and $\|a_1\| = \|a_2\|$, $\|b_1\| = \|b_2\|$. Let $H_a : [0, 1] \times [0, \|a\|] \rightarrow X$ and $H_b : [0, 1] \times [0, \|b\|] \rightarrow X$ be the corresponding homotopies:

$$H_a(0, t) = a_1(t), \quad H_a(1, t) = a_2(t), \quad H_b(0, t) = b_1(t), \quad H_b(1, t) = b_2(t).$$

Define $H : [0, 1] \times [0, \|a\| + \|b\|] \rightarrow X$ by

$$H(s, t) = \begin{cases} H_a(s, t), & 0 \leq t \leq \|a\|, \\ H_b(s, t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|. \end{cases}$$

The map H is continuous by the pasting lemma, since $H_a(s, \|a\|) = a_1(\|a\|) = b_1(0) = H_b(s, 0)$ for all s . Then H is a homotopy between $a_1 \cdot b_1$ and $a_2 \cdot b_2$, proving that composition is well-defined on homotopy classes.

(2) Strict Associativity. Let $a : [0, \|a\|] \rightarrow X$, $b : [0, \|b\|] \rightarrow X$, $c : [0, \|c\|] \rightarrow X$ be composable paths with $a(\|a\|) = b(0)$ and $b(\|b\|) = c(0)$. Define their concatenations

$$(a \cdot b) \cdot c : [0, \|a\| + \|b\| + \|c\|] \rightarrow X, \quad a \cdot (b \cdot c) : [0, \|a\| + \|b\| + \|c\|] \rightarrow X.$$

By the definition of path concatenation, both are the same function on $[0, \|a\| + \|b\| + \|c\|]$:

$$((a \cdot b) \cdot c)(t) = a \cdot (b \cdot c)(t) = \begin{cases} a(t), & 0 \leq t \leq \|a\|, \\ b(t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|, \\ c(t - \|a\| - \|b\|), & \|a\| + \|b\| \leq t \leq \|a\| + \|b\| + \|c\|. \end{cases}$$

Thus $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ as functions, making composition strictly associative on the groupoid level.

(3) Identities. For each $p \in X$, let $e_p : [0, 0] \rightarrow X$ be the identity path with $e_p(0) = p$ (defined on the singleton domain). For any $a : [0, \|a\|] \rightarrow X$ with $a(0) = p$ and $a(\|a\|) = q$, the concatenations are:

$$(e_p \cdot a)(t) = \begin{cases} e_p(t) = p, & 0 \leq t \leq 0, \\ a(t), & 0 \leq t \leq \|a\|, \end{cases} = a(t) \quad \text{for all } t \in [0, \|a\|].$$

Similarly, $(a \cdot e_q)(t) = a(t)$ for all $t \in [0, \|a\|]$. Thus $[e_p]$ acts as a left and right identity in $\Gamma(X)$.

(4) Inverses. For $a : [0, \|a\|] \rightarrow X$ with $a(0) = p$ and $a(\|a\|) = q$, define $a^{-1} : [0, \|a\|] \rightarrow X$ by $a^{-1}(t) = a(\|a\| - t)$. Then $a^{-1}(0) = a(\|a\|) = q$ and $a^{-1}(\|a\|) = a(0) = p$, so $a^{-1} : \Gamma(X)(q, p)$.

We must show that $a \cdot a^{-1} \simeq e_q$. Define a homotopy $H : [0, 1] \times [0, 2\|a\|] \rightarrow X$ by

$$H(s, t) = \begin{cases} a(t), & 0 \leq t \leq (1-s)\|a\|, \\ a((1-s)\|a\|), & (1-s)\|a\| \leq t \leq (1+s)\|a\|, \\ a(\|a\| - (t - (1+s)\|a\|)), & (1+s)\|a\| \leq t \leq 2\|a\|. \end{cases}$$

At $s = 0$ this gives $a \cdot a^{-1}$, and at $s = 1$ the map collapses to the constant path e_q . Hence $a \cdot a^{-1} \simeq e_q$, and by symmetry $a^{-1} \cdot a \simeq e_p$. Passing to homotopy classes yields $[a] \cdot [a^{-1}] = [e_q]$ and $[a^{-1}] \cdot [a] = [e_p]$, completing the proof. $\square$

Basic Properties

Theorem 2.4.3. *For any path $a : [0, \|a\|] \rightarrow X$, there exist identity paths $e_1 : [0, 0] \rightarrow X$*

and $e_2 : [0, 0] \rightarrow X$ such that $a \cdot a^{-1} \simeq e_1$ and $a^{-1} \cdot a \simeq e_2$, where $\simeq$ denotes homotopy relative to endpoints.

Proof. Let $p = a(0)$ and $q = a(\|a\|)$, so $e_1(0) = q$ and $e_2(0) = p$. Define $H : [0, 1] \times [0, 2\|a\|] \rightarrow X$ by

$$H(s, t) = \begin{cases} a(t), & 0 \leq t \leq (1-s)\|a\|, \\ a((1-s)\|a\|), & (1-s)\|a\| \leq t \leq (1+s)\|a\|, \\ a(\|a\| - (t - (1+s)\|a\|)), & (1+s)\|a\| \leq t \leq 2\|a\|. \end{cases}$$

Then $H(0, t) = (a \cdot a^{-1})(t)$ for $t \in [0, 2\|a\|]$ and $H(1, t) = q$ for all $t \in [0, 2\|a\|]$. Similarly, by symmetry, we can construct a homotopy collapsing $a^{-1} \cdot a$ to e_2 . Thus both concatenations are homotopic to their respective identity paths. $\square$

Theorem 2.4.4. *If $a \simeq b$ (with the same stopping time), then $a^{-1} \simeq b^{-1}$.*

Proof. Let $H : [0, 1] \times [0, \|a\|] \rightarrow X$ be a homotopy from a to b . Define $\tilde{H}(s, t) = H(s, \|a\| - t)$. Then $\tilde{H}(0, t) = a^{-1}(t)$, $\tilde{H}(1, t) = b^{-1}(t)$, and for all s ,

$$\tilde{H}(s, 0) = H(s, \|a\|) = a(\|a\|) = b(\|a\|), \quad \tilde{H}(s, \|a\|) = H(s, 0) = a(0) = b(0).$$

Hence $\tilde{H}$ is a homotopy between a^{-1} and b^{-1} relative to endpoints. $\square$

Theorem 2.4.5. *For any $\alpha \in \Gamma(X)(p, q)$, there exist identities $\epsilon_q : [0, 0] \rightarrow X$ and $\epsilon_p : [0, 0] \rightarrow X$ such that $\alpha \cdot \alpha^{-1} = \epsilon_q$ and $\alpha^{-1} \cdot \alpha = \epsilon_p$.*

Proof. Let $\alpha = [a]$ where $a : [0, \|a\|] \rightarrow X$ with $a(0) = p$ and $a(\|a\|) = q$. By Theorem 2.4.3, we have $a \cdot a^{-1} \simeq e_q$ and $a^{-1} \cdot a \simeq e_p$, where $e_q : [0, 0] \rightarrow X$ with $e_q(0) = q$ and $e_p : [0, 0] \rightarrow X$ with $e_p(0) = p$. Passing to homotopy classes yields

$$\alpha \cdot \alpha^{-1} = [a \cdot a^{-1}] = [e_q] =: \epsilon_q, \quad \alpha^{-1} \cdot \alpha = [a^{-1} \cdot a] = [e_p] =: \epsilon_p.$$

Thus each morphism has a two-sided inverse in $\Gamma(X)$. $\square$

2.5 The Fundamental Group

Definition 2.5.1. Let X be a topological space and $p \in X$ a fixed basepoint. A *loop at p* is a path $a : [0, \|a\|] \rightarrow X$ such that $a(0) = a(\|a\|) = p$.

Two loops a, b at p with the same stopping time $\|a\| = \|b\|$ are said to be *homotopic relative to endpoints* if there exists a continuous map

$$H : [0, 1] \times [0, \|a\|] \rightarrow X$$

such that

$$H(0, t) = a(t), \quad H(1, t) = b(t), \quad H(s, 0) = H(s, \|a\|) = p \quad \forall s, t \in [0, \|a\|].$$

The set of all such homotopy classes of loops at p is denoted by

$$\pi_1(X, p) = \{[a] \mid a(0) = a(\|a\|) = p\},$$

and is called the *fundamental group of X at p* . The group operation is defined by *path concatenation*:

$$[a] \cdot [b] = [a * b],$$

where

$$(a * b)(t) = \begin{cases} a(t), & 0 \leq t \leq \|a\|, \\ b(t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|. \end{cases}$$

The result $a * b$ is a loop with stopping time $\|a * b\| = \|a\| + \|b\|$.

Theorem 2.5.2. *The set $\pi_1(X, p)$, together with the operation $[a] \cdot [b] = [a * b]$, forms a group.*

Proof. We verify the four group axioms: closure, associativity, identity, and inverses.

(1) Closure. If a, b are loops at p , then $a(\|a\|) = p = b(0)$, and hence $a * b$ is continuous by the pasting lemma. Moreover,

$$(a * b)(0) = a(0) = p, \quad (a * b)(\|a\| + \|b\|) = b(\|b\|) = p,$$

so $a * b$ is again a loop at p . To show this operation is well-defined on homotopy classes,

suppose $a \simeq a'$ and $b \simeq b'$ relative to endpoints, with $\|a\| = \|a'\|$ and $\|b\| = \|b'\|$. Then the concatenated homotopy

$$H(s, t) = \begin{cases} H_a(s, t), & 0 \leq t \leq \|a\|, \\ H_b(s, t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|, \end{cases}$$

where H_a, H_b are homotopies between a, a' and b, b' , shows $a * b \simeq a' * b'$. Thus the operation is well-defined.

(2) Associativity. For loops a, b, c at p , the paths $(a * b) * c$ and $a * (b * c)$ have the same domain $[0, \|a\| + \|b\| + \|c\|]$ and are the same function:

$$((a * b) * c)(t) = a * (b * c)(t) = \begin{cases} a(t), & 0 \leq t \leq \|a\|, \\ b(t - \|a\|), & \|a\| \leq t \leq \|a\| + \|b\|, \\ c(t - \|a\| - \|b\|), & \|a\| + \|b\| \leq t \leq \|a\| + \|b\| + \|c\|. \end{cases}$$

Thus $(a * b) * c = a * (b * c)$ exactly, so associativity holds in $\pi_1(X, p)$ without any need for homotopy.

(3) Identity. Let $e_p : [0, 0] \rightarrow X$ be the identity loop at p , with $e_p(0) = p$ (the constant loop on the singleton domain). For any loop $a : [0, \|a\|] \rightarrow X$ at p , we have

$$(e_p * a)(t) = \begin{cases} e_p(t) = p, & 0 \leq t \leq 0, \\ a(t), & 0 \leq t \leq \|a\|, \end{cases} = a(t) \quad \text{for all } t \in [0, \|a\|].$$

Thus $(e_p * a) = a$ and by the same argument $(a * e_p) = a$, so $[e_p]$ serves as the identity element.

(4) Inverses. For a loop $a : [0, \|a\|] \rightarrow X$ at p , define its inverse $a^{-1} : [0, \|a\|] \rightarrow X$ by $a^{-1}(t) = a(\|a\| - t)$. Then $a^{-1}(0) = a(\|a\|) = p$ and $a^{-1}(\|a\|) = a(0) = p$, so a^{-1} is

also a loop at p . The concatenation $a * a^{-1} : [0, 2\|a\|] \rightarrow X$ is homotopic to e_p via

$$H(s, t) = \begin{cases} a(t), & 0 \leq t \leq (1-s)\|a\|, \\ a((1-s)\|a\|), & (1-s)\|a\| \leq t \leq (1+s)\|a\|, \\ a(\|a\| - (t - (1+s)\|a\|)), & (1+s)\|a\| \leq t \leq 2\|a\|. \end{cases}$$

At $s = 0$ we have $(a * a^{-1})(t)$ for $t \in [0, 2\|a\|]$, and at $s = 1$ the path collapses to the constant map equal to p . Thus $[a * a^{-1}] = [e_p]$ in $\pi_1(X, p)$. Similarly, $[a^{-1} * a] = [e_p]$. Hence $[a^{-1}]$ is the inverse of $[a]$.

Since closure, associativity, identity, and inverses all hold, $\pi_1(X, p)$ is a group. $\square$

2.6 Change of Basepoint

Definition 2.6.1. A space X is *pathwise connected* if for any $p, q \in X$, there exists a path $c : [0, \|c\|] \rightarrow X$ (with some $\|c\| > 0$) such that $c(0) = p$ and $c(\|c\|) = q$.

Theorem 2.6.2. If X is pathwise connected and $c : [0, \|c\|] \rightarrow X$ is a path from p to q , then the map

$$\Phi_c : \pi_1(X, p) \rightarrow \pi_1(X, q), \quad \Phi_c([a]) = [c^{-1} \cdot a \cdot c]$$

is an isomorphism.

Proof. We must verify that Φ_c is well-defined, a homomorphism, and has an inverse.

Well-definedness: If $a \simeq a'$ relative to endpoints with $\|a\| = \|a'\|$, then the three paths $c^{-1} \cdot a \cdot c$ and $c^{-1} \cdot a' \cdot c$ have the same domain $[0, \|c\| + \|a\| + \|c\|]$ and the homotopy between a and a' induces a homotopy between the concatenations (using the pasting lemma). Thus $\Phi_c([a]) = \Phi_c([a'])$.

Homomorphism: For loops $[a], [b] \in \pi_1(X, p)$ with $\|a\|$ and $\|b\|$,

$$\Phi_c([a] \cdot [b]) = \Phi_c([a * b]) = [c^{-1} \cdot (a * b) \cdot c].$$

We compute the stopping time:

$$\|c^{-1} \cdot (a * b) \cdot c\| = \|c\| + (\|a\| + \|b\|) + \|c\| = \|c\| + \|a\| + \|c\| + \|b\|.$$

On the other hand,

$$[c^{-1} \cdot a \cdot c] \cdot [c^{-1} \cdot b \cdot c] = [(c^{-1} \cdot a \cdot c) * (c^{-1} \cdot b \cdot c)].$$

We need to verify that these are homotopic. Since $(c^{-1} \cdot a \cdot c)(\|c^{-1}\| + \|a\| + \|c\|) = c^{-1}(\|c\|) = c(0) = p$ and $(c^{-1} \cdot b \cdot c)(0) = c^{-1}(0) = c(\|c\|) = q$, the concatenation is not directly valid in the sense that the endpoint of the first does not equal the startpoint of the second.

Actually, we should reconsider: in $\pi_1(X, p)$, all loops are based at p , so for $[a], [b] \in \pi_1(X, p)$, both have $a(0) = a(\|a\|) = p$ and $b(0) = b(\|b\|) = p$. So the concatenation $a * b$ is valid and gives a loop with stopping time $\|a\| + \|b\|$.

Now $\Phi_c([a] \cdot [b]) = [c^{-1} \cdot (a * b) \cdot c]$, where $(a * b)$ is defined as before, giving a loop at p .

Also, $\Phi_c([a]) \cdot \Phi_c([b]) = [c^{-1} \cdot a \cdot c] * [c^{-1} \cdot b \cdot c]$.

Both $c^{-1} \cdot a \cdot c$ and $c^{-1} \cdot b \cdot c$ are loops at q (since $c^{-1}(0) = q$, $a(0) = p$, $c(\|c\|) = q$, so $(c^{-1} \cdot a \cdot c)(0) = c^{-1}(0) = q$ and $(c^{-1} \cdot a \cdot c)(\|c^{-1}\| + \|a\| + \|c\|) = c(\|c\|) = q$; similarly for $c^{-1} \cdot b \cdot c$). Thus their concatenation is valid.

We have

$$(c^{-1} \cdot a \cdot c) * (c^{-1} \cdot b \cdot c) = c^{-1} \cdot a \cdot c \cdot c^{-1} \cdot b \cdot c.$$

Since $c \cdot c^{-1} \simeq e_q$ (the identity loop at q), and post-composition by homotopy is compatible with path concatenation, we have

$$c^{-1} \cdot a \cdot (c \cdot c^{-1}) \cdot b \cdot c \simeq c^{-1} \cdot a \cdot e_q \cdot b \cdot c = c^{-1} \cdot a \cdot b \cdot c = c^{-1} \cdot (a * b) \cdot c.$$

Thus Φ_c is a homomorphism.

Inverse: The inverse map is $\Phi_{c^{-1}} : [c] \circ \circ [c^{-1}] : \pi_1(X, q) \rightarrow \pi_1(X, p)$ given by $\Phi_{c^{-1}}([a]) = [c \cdot a \cdot c^{-1}]$.

Indeed, $\Phi_c \circ \Phi_{c^{-1}}([a]) = \Phi_c([c \cdot a \cdot c^{-1}]) = [c^{-1} \cdot (c \cdot a \cdot c^{-1}) \cdot c]$.

Since $c^{-1} \cdot c \simeq e_p$, we have

$$c^{-1} \cdot c \cdot a \cdot c^{-1} \cdot c \simeq e_p \cdot a \cdot e_p = a,$$

so $\Phi_c \circ \Phi_{c^{-1}} = \text{id}_{\pi_1(X, q)}$. Similarly, $\Phi_{c^{-1}} \circ \Phi_c = \text{id}_{\pi_1(X, p)}$.

Thus Φ_c is an isomorphism. □

Remark 2.6.3. Thus, in a pathwise connected space, all fundamental groups based at different points are naturally isomorphic. We may therefore refer to $\pi_1(X)$ without specifying a basepoint.

Chapter 3

Lattice Knots

3.1 Definitions

Definition 3.1.1. Lattice knots or cubulated knots in $3 - D$ are polygonal knots whose sides are parallel to one of the axes .

Theorem 3.1.2 (Cubulation). *Every tame knot in $\mathbb{R}^3$ is isotopic to a lattice knot.*

Definition 3.1.3. A lattice knot may be encoded as a reduced word in the alphabet

$$\{N, S, E, W, U, D\}.$$

We shall call this representation a *knotword*. For instance, an unknot is given by the knotwords **NESW**, and **NNESUWWDDDEESUW**.

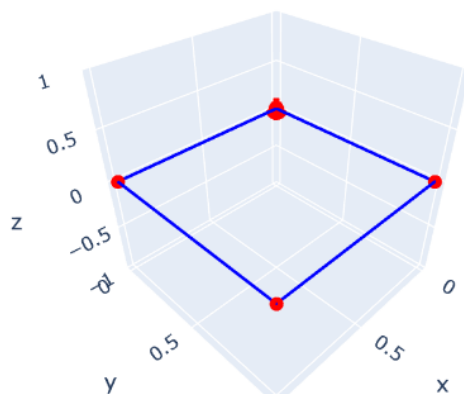


Figure 3.1: The knot NESW

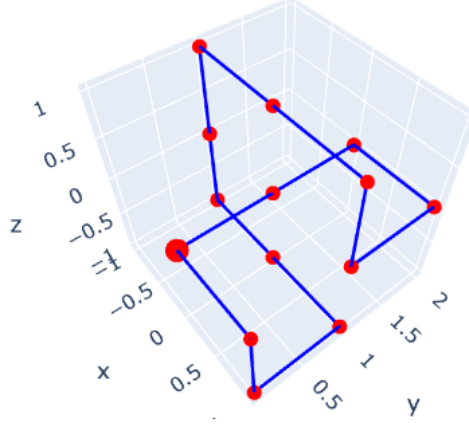


Figure 3.2: The knot NNESUWWDDDEESUW

For a word w , we define its displacement in $\mathbb{R}^3$ as:

$$D(w) = (|w|_E - |w|_W, |w|_N - |w|_S, |w|_U - |w|_D).$$

For a word $w = w_1 \dots w_{|w|}$, let us define the subword $w[i : j]$ for $1 \leq i \leq j \leq |w|$ as the contiguous segment $w_i \dots w_j$.

Further, let us call a subword $w' = w[i : j]$ *proper* iff $w' \neq w$ and $w' \neq \epsilon$.

It is useful to note the additive property of D over concatenation:

$$D(ww') = D(w) + D(w')$$

In particular, it follows that for a substring w_{sub} of knotword $w = w_1 w_{sub} w_2$, the substring-deleted word $w' := w_1 w_2$ obeys

$$D(w') = D(w) - D(w_{sub})$$

Of course, $D(\epsilon) = 0$.

Definition 3.1.4. A nontrivial word w represents a lattice knot if:

1. $D(w) = \mathbf{0}$,
2. no proper subword w' has $D(w') = \mathbf{0}$.

The first criterion means that the curve is closed and the second means that the curve does not have any self-intersection.

3.2 Elementary Moves

Definition 3.2.1 (Elementary Switch). Let v be a subword of w . Replace v by NvS , SvN , or analogous substitutions using E and W , or U and D . Then reduce the string, meaning, recursively eliminate from the result any consecutive alphabets AB having $D(AB) = 0$. If the resulting word is still a lattice knot, then we call this operation an elementary switch.

Definition 3.2.2 (Doubling). Doubling in the x -direction is the application of substitutions $E \mapsto EE$ along with $W \mapsto WW$ (and analogously for y, z) applied to all occurrences of those letters.

Sasha Anan'in, Alexandre Grishkov, and Dmitrii Korshunov have proven the following two theorems

Theorem 3.2.3 (Doubling From Elementary Switches). *Let w be a lattice knot. Doubling w in any coordinate direction can be obtained by a finite sequence of elementary switches.*

Theorem 3.2.4 (Isotopy). *Two lattice knots w_1 and w_2 are isotopic if and only if one can be transformed into the other by a finite sequence of elementary switches.*

We can generate all reduced words of length $2n$ satisfying conditions (1) and (2) in definition (4.1.4). The shortest nontrivial lattice knot occurs at length 24 and the next new nontrivial lattice knot has length 30. But even enumerating all reduced words of length 50 is computationally very expensive. In case we are given two valid knot diagrams, we can check for all the possible sequences of elementary switches of length k . But this method has exponential time complexity and is infeasible for almost all practical cases. No polynomial-time algorithm is known for this.

n	Number of valid knot sequences of length $2n$
1	0
2	24
3	264
4	3312
5	48240
6	762096
7	12,673,920
8	218,904,768
9	3,891,176,352
10	70,742,410,800
11	1,309,643,747,808
12	24,609,869,536,800
13	468,270,744,898,944
14	9,005,391,024,862,848
15	174,776,445,357,365,040
16	3,419,171,337,633,496,704

Table 3.1: Number of valid knot sequences for words of length $2n$.

Chapter 4

Questions from knot theory

4.1 Planar Knots Must Be the Unknot

We are given a knot $K \subseteq \mathbb{R}^2$, which is the image of an imbedding, say $\kappa : S^1 \mapsto \mathbb{R}^2$. Let us import a well known result of point-set topology, which may be found in [2].

Theorem 4.1.1 (Jordan–Schoenflies). *If $C \subset \mathbb{R}^2$ is the image of an embedding $S^1 \hookrightarrow \mathbb{R}^2$, then $\mathbb{R}^2 \setminus C$ has exactly two connected components: one bounded (the interior) and one unbounded (the exterior). Let D denote the bounded component of $\mathbb{R}^2 \setminus C$. Then there exists a homeomorphism*

$$\varphi : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$$

such that

$$\varphi(C) = S^1, \quad \varphi(D) = \{x \in \mathbb{R}^2 : \|x\| < 1\}.$$

Let us obtain φ_K by applying Jordan–Schoenflies taking $C = K$. $\text{Dom}(\varphi_K)$ extends to 3D, given by

$$H = (x, y, z) \mapsto (\Pi_X \varphi_K(x, y), \Pi_Y \varphi_K(x, y), z)$$

where Π_X and Π_Y are coordinate projections. Now, notice that H is the product of two homeomorphisms, namely

$$H_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2 = (x, y) \mapsto \varphi(x, y)$$

$$H_2 : \mathbb{R} \rightarrow \mathbb{R} = z \mapsto z$$

Thus $H_1 \times H_2 : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^2 \times \mathbb{R}$ is a homeomorphism, proving that the product map, namely H , is an ambient homeomorphism.

Let us identify $\mathbb{R}^2$ with the XY-plane. It is clear from this construction that $H(K) = S^1 \subseteq \mathbb{R}^2$, and we have found our ambient homeomorphism. Thus, by definition K is equivalent to the unknot.

—

4.2 Countability of Tame Knot Types

Let T be the set of equivalence classes of ambiently isotopic tame knots. Any $K \in [K] \in T$ has, by virtue of tameness, has a 3D polygonal representation, say P . By density of the rational numbers, we can cause (via an ambient isotopy) arbitrarily small perturbations into each of the vertices of P to move them to rational coordinates. That is to say,

Proposition 4.2.1. *Every polygon in 3D is ambient isotopic to a rational polygon under arbitrarily small spatial displacements.*

We proceed with the same arguments as formalized in Chapter 1, Section 3.1, Page 7 of [3] for the specific case of arbitrarily small rotations of the polygon.

We see that $|T| = \aleph_0$ follows from the countability of $\mathbb{Q}^3$.

—

4.3 Dowker Parity lemma for regular projection of knot

Dowker notation is a concise way to describe a knot projection using integers. To construct it, choose an orientation on a knot diagram and start at any crossing, labelling it 1. Follow the orientation along the *understrand* to the next crossing and label it 2, then continue around the knot, assigning labels 3, 4, 5, ... in order until you return to the starting point. Since the knot passes through each crossing twice, every crossing receives exactly two labels—one from each time the strand passes through. An important observation is that every crossing ends up with one odd label and one even label. *Why must every crossing receive exactly one odd-numbered label and one even-numbered label?*

Answer. We would use that fact that every regular projection of knot has a checkerboard coloring. Suppose a crossing of the diagram receives the two labels n and m with $n < m$. Start at the crossing at the moment it receives the label n , and traverse the knot once in the chosen orientation until the second visit to that same crossing, which receives the label m . Let r denote the number of times we encounter a crossing during this .

If we discard parts of the diagram not encountered during this traversal and join the string at the crossing with label n and m , we obtain a closed curve in the plane whose crossings are visited during the traversal exactly twice. Say there are p such crossings. Apply a checkerboard coloring of the regions of the plane determined by this reduced diagram. Now $r = 2p + q$ where q is the number of crossing between the new knot and part of the original knot not traversed. when the curve enters and exits the new ,knot it enters through and exits to the same color. We know color of the changes when the curve crosses a strand of the reduced knot diagram (assuming the curve does not pass through a crossing). So the color changes even number of times, so the curve crosses the reduced knot diagram an even number of times . Thus the number q and r are also even.

$$m = n + r + 1 = n + 2k + 1,$$

so $m - n$ is odd. In other words, each crossing receives one odd and one even label.

4.4 Lattice Knots: Length Bounds for non-trivial knots

Our goal here is to find the lower bound for the length of a word which represents a non trivial knot . We will prove that this value will be at least 20. For that let us define **Minimal bounding box** which is the dimension (a, b, c) of the minimal box which can bound a given knot. Any non trivial lattice knot cannot be bound by a $(1, 1, n); n \in \mathbb{N}$ because it has to be the surface of the cuboid, which resolves to the unknot. We can also show it cannot be bounded by a $(2, 2, 2)$ box. Let us consider the part of the knot in the interior of the box. If there is no such part, then the knot wraps around the surface of a cube and resolves to the unknot. Let us handle the case when this interior part exists. It consists of two line segments and falls into one of only two cases:

1. a line through the center of two opposite faces

2. union of two lines from center of two adjacent faces to the center of the cube.

In both cases, the part of the knot on the surface can be modified so that the knot is an unknot because *any two simple polygonal arcs in S^2 with the same endpoints are ambient isotopic relative to the endpoints*.

Now, non-boundedness in $(2, 2, 2)$ obviously implies that it cannot be bounded by $(1, 2, 2)$. Next, assume a bounding box of size $(1, 2, 3)$. If a knot K bounded by this is non-trivial then it must contain both the interior edges, otherwise, it falls into one of the cases covered before and thus will be an unknot. Further assume K has the minimal length. Now look

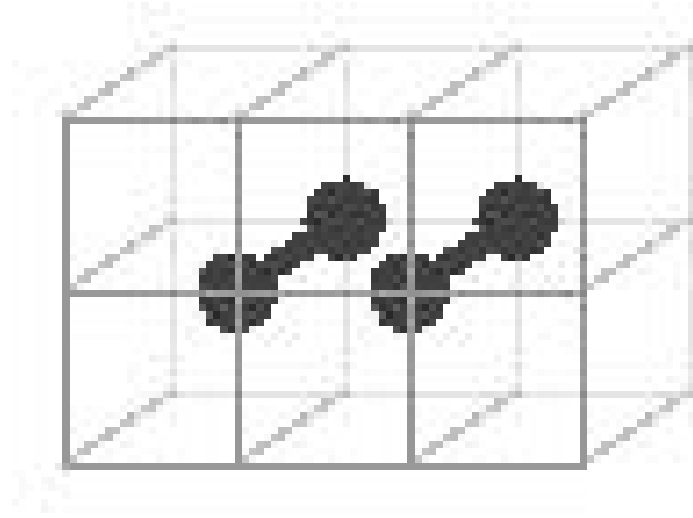


Figure 4.1: Intersection of K with interior of the box

at the intersection J of the knot with the near side of the box, i.e., the face nearer to the camera in the picture Figure 4.1. J cannot have length 1 as then it will have a u-turn-like structure so we can move this u-turn to a direct edge (Notice it cannot intersect with the knot during this movement). This is knot possible as K is minimal. Similarly we can show that J also cannot have size two by enumerating such cases.

So J can have length at least 3, which means they can look like any of the cases in Figure 4.3. All of these cases can be manually verified to wind up as an unknot or to get us a link. So if the minimal bounding box B of a non trivial knot is (a, b, c) , then we have exhausted all such triples that sum to 6 or less. We have thus shown $7 \leq (a + b + c)$.

Now we prove that a knot bounded by the minimal bounding box B has length greater than or equal to $2(a + b + c) + 6$.

The range of K along the x-axis is a . We claim the knot has length at least $2a$ along the x-axis, by virtue of being a closed loop.

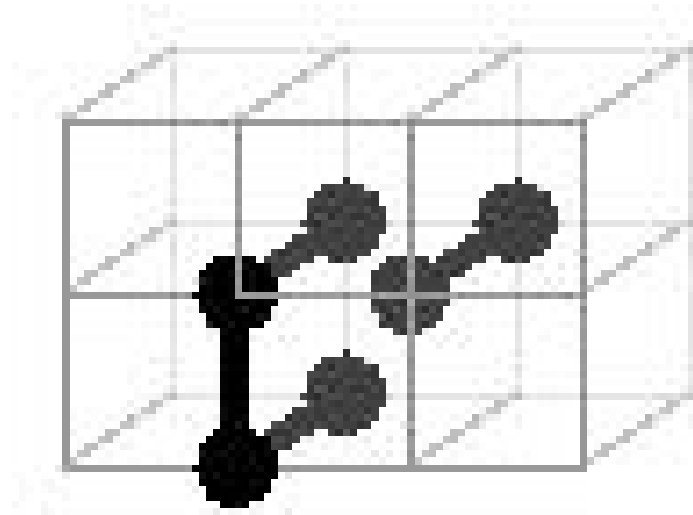
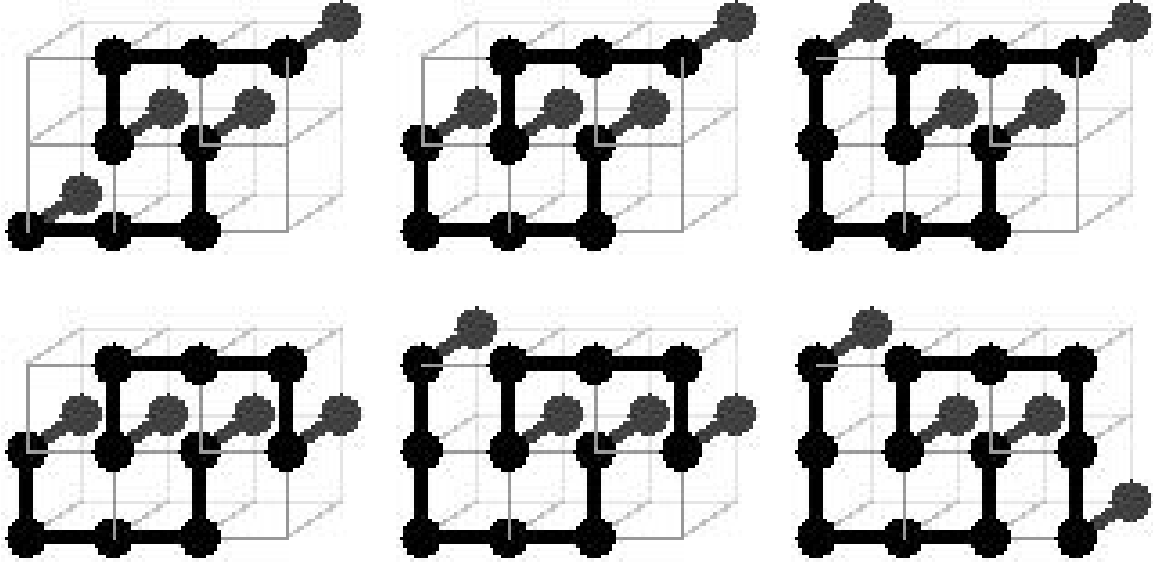


Figure 4.2: If length of J is 1

Figure 4.3: Possible cases for length of $J \geq 3$



Proposition 4.4.1. *Let K be a cubulated knot with knotword w . If the range of K along the X -axis is a , then $|w|_E + |w|_W \geq 2a$. Analogous statements follow for Y , and Z .*

Lemma 4.4.1.1 : Given $a \in \mathbb{Z}_{>0}$ and an integer sequence $d = (d_i)_{i=0}^n$ obeying:

1. For all $1 \leq i \leq n : |d(i) - d(i-1)| = 1$
2. There exist $0 \leq i \leq j \leq n$ such that $|d(j) - d(i)| = a$

Then, we must have $j - i \geq a$.

Pf: $a = |d(j) - d(i)| \leq \sum_{k=i+1}^j |d(k) - d(k-1)| = j - i$

Proof 4.4.1: Let us eliminate the N, S, U, D from w and call the resulting word w' .

Consider the integer sequence $d = (D(w'[1 : i]) \cdot \mathbf{e}_1)_{i=0}^{|w'|}$. This sequence by definition

has the property, for $1 \leq i \leq |w'|$:

$$d(i) - d(i-1) = D(w'[1 : i]) \cdot \mathbf{e}_1 - D(w'[1 : i-1]) \cdot \mathbf{e}_1 \quad (4.1)$$

$$= D(w'[i : i]) \cdot \mathbf{e}_1 \quad (\text{substring subtractivity}) \quad (4.2)$$

$$= D(w'_i) \cdot \mathbf{e}_1 \quad (\text{singleton substring}) \quad (4.3)$$

$$= \begin{cases} +1, & \text{if } w'_i = E, \\ -1 & \text{if } w'_i = W. \end{cases} \quad (4.4)$$

So $|d(i) - d(i-1)| = 1$.

Now range of w' being $a \implies \exists j \geq 1 : d(j) = a$

Of course, $d(0) = D(\epsilon) \cdot \mathbf{e}_1 = 0$

And since w was a knotword, $d(|w'|) = D(w) \cdot \mathbf{e}_1 = 0$

Our lemma applied to $d(0), d(j)$ gives

$$j - 0 \geq |d(j) - d(0)| = a$$

Our lemma applied to $d(j), d(|w'|)$ gives

$$|w'| - j \geq |d(|w'|) - d(j)| = a$$

Adding gives $|w'| \geq 2a$.

Let's now show that if this value is exactly $2a$ that means if one starts at a point with minimum x-coordinate then the displacement function along the x-axis will be a function which is non-decreasing at first until it reaches the maximum value a , and then it decreases to 0.

We prove by induction on a that any such lattice knot is an unknot.

Base case: $a = 1$. In this case the knot has exactly two edges parallel to the x -axis, All remaining edges lie in the two planes $x = 0$ and $x = 1$ say AB and CD . Say A and B lie on $x = 0$. In each of these planes, the intersection of the knot with the plane is a simple arc connecting the two endpoints A and B. Since any two simple polygonal arcs in S^2 (or $\mathbb{R}^2$) with the same endpoints are ambient isotopic if the endpoints are fixed, each planar arc is isotopic to the straight segment joining its endpoints. Therefore, the entire knot is

isotopic to a single rectangle, hence to the unknot.

Inductive step. Assume the statement holds for $a - 1$.

Examine the plane $x = a - 1$. There are exactly two points P and Q where the edges parallel to the x -axis connecting $x = a - 2$ to $x = a - 1$ and $x = a - 1$ to $x = a$ intersect.

Now consider the intersection of the knot with the plane $x = a$. This intersection is a simple arc connecting two endpoints lying on that plane. Since any two simple arcs in a plane with fixed endpoints are ambient isotopic, we may choose *within the plane* $x = a$ a simple polygonal arc connecting these endpoints whose projection to the yz -plane does not intersect the projection of the portion of the knot lying in the plane $x = a - 1$.

This is possible because a finite collection of line segments in a plane cannot disconnect the plane unless they form a closed curve; in our situation the knot is not entirely contained in the yz -plane, and it cannot have any self-intersections, so such a separating closed curve cannot occur.

Having chosen such an arc in $x = a$, we now shorten the two x -parallel edges that join the planes $x = a - 1$ and $x = a$, continuously sliding the planar arc from $x = a$ toward $x = a - 1$. Since this deformation does not introduce intersections, we obtain an ambient isotopy that pushes the entire portion of the knot lying in $x = a$ into the plane $x = a - 1$.

Thus, after the deformation, the knot lies entirely in the slab $0 \leq x \leq a - 1$ and uses only $2(a - 1)$ edges parallel to the x -axis. The resulting knot is therefore, by the inductive hypothesis, isotopic to the unknot.

So length along x -axis is at least $2a + 2$ for non trivial knot. Applying the same result for all the axes, we can prove our claim. Put $a + b + c \geq 7$ and we will get our lower bound of 20.

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